Big Data and AI in Marketing

AI Agent for Marketing Group Assignment

Option 2: **Spec-Driven Development with Claude Code**

Campaign Planning Agent

Group 1

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Table of Contents

1. Executive Summary

The Campaign Planning Agent is an AI-powered decision-support system designed to automate the marketing campaign planning process. Developed using a spec-driven methodology with Claude Code, the agent accepts between six and nine marketing inputs and executes a structured six-step workflow: classifying the campaign funnel stage, selecting appropriate marketing channels, allocating budgets, calculating key performance indicators (KPIs), generating relevant keywords, and producing a complete campaign brief. By automating these planning activities, the agent reduces a process that traditionally requires several hours of manual effort to just a few minutes.

To demonstrate its capabilities, the agent was applied to GlowDrop, a business-to-consumer (B2C) skincare brand seeking to increase first-time purchases. Based on the provided inputs, the agent generated a fully structured conversion-stage campaign brief, including channel recommendations, budget allocations, KPI targets, and campaign messaging. The resulting business value is both immediate and measurable, as the reduction in planning time translates directly into productivity gains and cost savings for marketing teams.

2. The Business Problem

Campaign planning is one of the most resource-intensive activities in marketing. For every new campaign, marketers must determine which channels to use, allocate budgets, establish KPIs such as Cost Per Acquisition (CPA), Click-Through Rate (CTR), and Return on Advertising Spend (ROAS), and translate these decisions into a campaign brief that is both actionable and realistic. Collectively, these tasks create a process that is often time-consuming and in fact, according to HubSpot's 2024 State of Marketing Report, marketers spend approximately four hours per day on manual tasks (Bodiroza, 2024).

This challenge is further compounded by the level of marketing expertise required – Effective campaign planning demands an understanding across multiple areas such as channel dynamics, audience behavior, performance benchmarks, and budget optimization. Smaller organizations that lack dedicated media planners or paid advertising specialists are often forced to make strategic decisions under tight time constraints and with incomplete knowledge or information. This often results in inconsistent campaign strategy, where two marketers working with the same data can recommend different channels, allocate budgets differently or select different KPI targets and overall leading to variations in the campaigning purpose and execution. These inconsistencies can delay campaign launches, create unrealistic performance expectations, and reduce alignment across marketing teams and platforms.

The fragmented nature of modern advertising ecosystems further amplifies this challenge as well. Marketers frequently need to coordinate campaigns across multiple platforms (e.g. Google Ads, Meta Ads and TikTok Ads), in which each have their own metrics, options, and pricing models. Consequently, manual planning not only increases the workload involved in campaign development but also raises the risks of errors and inefficiencies.

In the test example of GlowDrop, a B2C skincare brand with 42,000 followers on TikTok, despite its relatively strong brand awareness, the brand still struggles with converting this awareness into actual purchases through paid media. This highlights a gap between audience engagement and campaign execution, presenting a clear opportunity for AI-driven solutions to support campaign planning and decision-making.

3. Agent Concept Brief

3.1 Overview and the Marketing Problem Addressed

The Campaign Planning Agent is an AI-powered tool designed to automate the process of marketing campaign planning. Built using a spec-driven development approach, the agent takes a set of brand and campaign inputs provided to generate a campaign brief. This is done through going through a structured six-step pipeline, with the agent making a specific decision and justifying its decision based on the inputs provided, thereby ensuring a tailored campaign recommendation instead of a generic one. The output provides a ready-to-use campaign brief that teams can act on immediately or at minimum a structured baseline they can further refine, without additional research or manual planning required. As explained in Section 2, the agent directly addresses the time inefficiencies and expertise gaps that traditional campaign planning processes commonly face.

3.2 Target Users

The agent is designed for marketing teams across organizations of all sizes – from large corporations and agencies to small businesses and freelance marketers. It is particularly valuable for those who require structured campaign planning support but have limited time or specialist expertise, and for those who need to move quickly from brand context to a deployable campaign brief without spending hours on manual research and planning.

3.3 Overview of Inputs Needed

The AI Agent accepts 9 inputs from the users, of which 6 are mandatory:

No.	Input	Mandatory?	Description
1	Marketing Goal	Yes	What the brand wants to achieve
2	Brand Description	Yes	Context of the brand
3	Target Audience	Yes	Who the campaign aims to target
4	Total Budget	Yes	Total available spend
5	Product Price	Yes	Price of the product or service
6	Currency	No	Currency *(*Default: GBP)*
7	Target Market Language	Yes	Language of the target market
8	Existing Channels	No	Marketing channels the brand currently use
9	Product Type	No	Business-to-Business (B2B) or B2C

3.4 Core Workflow
The core workflow encompasses a six-step sequential pipeline as detailed below:

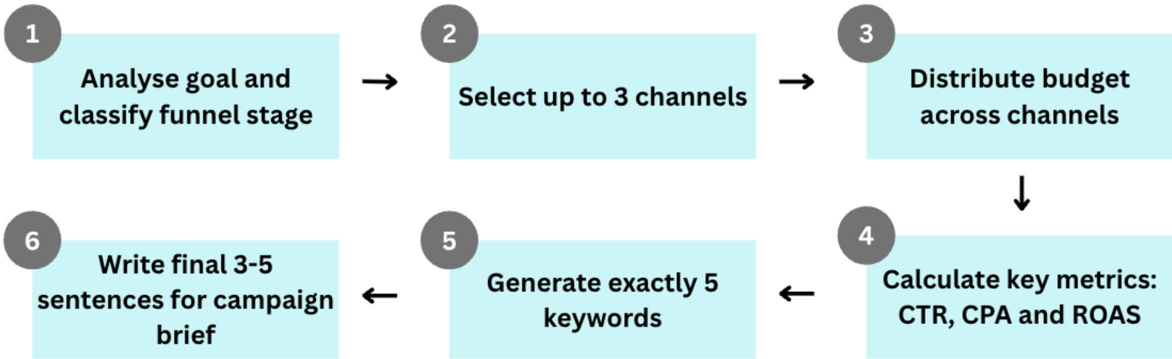

Figure 1: Six-step Pipeline of the Campaign Planning Agent

The system qualifies as an agent rather than a simple prompt wrapper because it has the ability to make decisions, provide explanation of its reasoning at each step and passes structured outputs from one step to another. A full demonstration of the agent running with a real input scenario is provided in Section 5.

3.5 Final Output
After completing the six-step pipeline, the agent produces a campaign package consisting of a recommended funnel stage with the following components:

- A recommended funnel stage with reasoning
- Up to three selected advertising channels with pricing models
- The budget allocation across the selected channels
- KPI targets (CTR, CPA and ROAS) of each selected channel
- Five keywords in the target market language
- Ready-to-use campaign brief in 3-5 sentences

The detailed output and walkthrough can be found in Section 5.

3.6 Expected Business Value
The AI Agent is expected to deliver significant time savings for marketing teams. Based on our test iterations, the agent is capable of compressing hours of manual work into around 2 minutes. This allows marketing teams to redirect their time to other high-value activities such as creative development. A detailed quantification of business impact is elaborated further in Section 6.

4. Workflow and Technical Design

4.1 Overview

The agent operates as a linear pipeline consisting of six steps, each responsible for making a specific decision and passing its output to the next step. This structure was specifically chosen to improve code readability and simplify debugging. Most notably, the output at each step is not simply just a prompt, but rather the agent goes through a reasoning process to explain the decision made.

4.2 The Six-Step Pipeline

Before the pipeline begins, the agent first receives a total of nine inputs (six mandatory, three optional) – refer to Section 3.3 for full description. Once the agent has received the required inputs, the six-step pipeline is initiated, with each step making a specific decision and passing its output to the next.

Step	Function	What it decides
Step 1	analyze_goal()	Reads the marketing goal and decides whether the campaign classifies as one of the following stages • Awareness • Consideration • Conversion stage The agent also provides a one-line explanation for the classification made.
Step 2	select_channels()	Selects up to 3 channels based on the funnel stage and audience. Each one comes back with a pricing model – Cost Per Click (CPC), Cost Per Mile (CPM) or Cost Per Acquisition (CPA), and a short justification.
Step 3	distribute_budget()	Splits the budget across the selected channels by priority. Assigns percentages and absolute amounts. The total is forced to add up to exactly 100% in Python, not just in the prompt.
Step 4	calculate_kpis()	Works out CTR, CPA, and ROAS targets from the actual product price and budget — not from generic benchmarks. Every CPA is checked to sit below the product price.
Step 5	generate_keywords()	Produces exactly 5 keywords matched to the funnel stage and the target market's language. Count and transactional intent are validated before the output moves on.
Step 6	write_brief()	Takes all the outputs from the five previous steps and writes a 3 to 5 sentence campaign brief that can go straight to a media buyer without editing.

The steps are connected sequentially – the funnel stage from Step 1 drives channel selection in Step 2, the selected channels feed into the budget allocation in Step 3, and the product price, budget and funnel stage inform the KPI targets and keywords in Steps 4 and 5. Step 6 concludes everything into the final campaign brief. The detailed walkthrough of the agent can be found in Section 5 with further details.

It is important to highlight that the budget allocation and KPI calculations are not left entirely to the AI model – budget, CPA and ROAS are calculated in Python to ensure that the numbers are always accurate regardless of the model's output. Each function is also called only with the data it requires at that particular stage, hence individual steps can be tested independently and modified in isolation without affecting the rest of the pipeline.

4.3 How We Built It

Since Option 2: spec-driven development with Claude Code was selected for the group assignment, the approach involved writing a complete specification describing what the software should do before any code is written and subsequently using that specification to drive all development decisions. A key advantage of this approach was that all inputs, outputs and constraints were defined upfront. As such, when the code was generated, the logic was already fully planned, and the expected behavior of each function was clearly defined.

The development followed six phases:

Phase	What we did	Prompt / command
Setup	Install Claude Code, log in, create the project folder.	npm install -g @anthropic-ai/claude-code → claude login → mkdir campaign-agent → claude
Spec	Write CLAUDE.md by hand with the 6 steps, constraints, and edge cases.	*Written by the team (not generated)*
Requirements	Ask Claude Code to formalize the spec into a requirements document.	Read my CLAUDE.md and produce a structured requirements document with all inputs, outputs, processing logic, constraints, and edge cases. Save as REQUIREMENTS.md.
Design	Ask Claude Code to produce a design doc. No code yet.	Based on REQUIREMENTS.md, create a design document with file structure, function descriptions, data flow, and implementation tasks. Save as DESIGN.md. Do not write any code yet.
Build	Implement one function at a time, checking each one before moving on.	Implement [function name]. Follow DESIGN.md. Do not implement any other functions yet. (repeated x6)
Integration	Connect everything and run end-to-end with real data.	All 6 functions done. Create main.py connecting them into a single pipeline. Test with GlowDrop input.

Each function was implemented individually rather than generating the entire agent at once. While the latter could have been faster, it would have made it significantly harder to identify logic errors at each stage. Implementing and verifying one function at a time also made the final integration step considerably more straightforward, as each component had already been tested in isolation before being connected.

4.4 File Structure

The project is hosted on GitHub (campaign-planning-agent) and is organized as follows:

File Folder/	What it does
campaign-planning-agent/	Root folder of the project. Contains all source files and documentation.
steps/	One .py file per pipeline step. Each function handles exactly one decision and nothing else.
models/	Shared typed data models used across all steps, so the data structure stays consistent end to end.
validators/	Validation logic that checks outputs between steps — for example, confirming the budget sums to 100%, and the keyword count is exactly 5.
deliverables-docs/	Documents and materials produced by the team for this project: written report, presentation deck, demonstration recording and final output of the agent.
agent.py	Orchestrator. Calls the 6 step functions in order and passes each output downstream.
main.py	Entry point. Collects user input, runs the full pipeline, and prints the final brief.
test_run.py	Used to run the agent end-to-end with the GlowDrop test case and verify the output.
CLAUDE.md	The spec we wrote by hand — defines the 6 steps, inputs, outputs, constraints, and edge cases.
DESIGN.md	Generated by Claude Code from the spec. Covers file structure, function signatures, and data flow.
REQUIREMENTS.md	Also generated by Claude Code. Formalizes inputs, outputs, 9 constraints, and 5 edge cases.
requirements.txt	Python dependencies (39 bytes — minimal, just the Anthropic SDK).
README.md	Overall introduction to the GitHub repository. Covers setup instructions, how to run the agent etc.

It is worth noting that the *validators/* directory is deliberately kept separate from the *steps/* directory as validation and decision-making are distinct processes. Combining them would reduce the readability and maintenance of each step function. The *requirements.txt* file is intentionally minimal and reflects the agent's lightweight dependencies. *test_run.py* was used to execute the GlowDrop example and verify the end-to-end correctness of the pipeline, with the full output presented in Section 5.

4.5 How Constraints Are Enforced

A key design decision was to enforce critical constraints at two levels: (1) within the prompt instructions sent to the AI model and (2) in the Python code after the response is received. This dual enforcement was necessary to avoid overreliance on the AI model for constraints where mathematical precision is critical. All validation logic resides in the validators/ directory. The following constraints are enforced in code:

Constraint	Explanation
Budget Total	All allocations are normalized in Python to guarantee they sum to exactly 100%, regardless of any rounding discrepancies in the model's output.
Maximum CPA	CPA targets are checked against the product price. If a CPA target exceeds the product price, the campaign would operate at a loss — this is caught and flagged before the output is passed forward.
ROAS calculation	ROAS is always computed in Python as product price divided by CPA target, never generated by the model, ensuring the formula is always correct.
Keyword count	Output is validated to confirm exactly five keywords were returned.
Channel limit	Number of selected channels is capped at a maximum of three.

4.6 Other Technical Decisions

Additional design considerations include:

Consideration	Explanation
Scope-based prompts	Each function passes to the model only the parameters it requires at that stage, preventing unnecessary context from influencing the model's decisions.
Standardized output types	All functions return a Python dataclass rather than a raw string, ensuring the next function always receives a consistent and predictable data structure regardless of how the model phrased its response.
Separate reasoning	Every prompt asks the model for a one-line explanation alongside its decision. This reasoning is saved with the output and included in the final campaign brief for transparency.
Function-level memory only	The agent does not maintain any state between steps. Each function receives precisely the inputs it needs and nothing more.

Overall, the agent completes the full six-step pipeline in less than two minutes and produces a final output structured in a ready-to-use format that can be handed directly to a media buyer without further editing.

5. Working Prototype

To demonstrate the capability of the Campaign Planning AI Agent, this section walks through the complete end-to-end workflow, using GlowDrop as the test example. The full video demonstration of the agent can be found in the deliverable-docs folder in the GitHub repository.

5.1 Sample Inputs

The marketer first needs to provide the agent with up to 9 inputs as shown in Figure 2 below.

```
==============================================================
  Campaign Planning Agent
==============================================================

Enter the details below. Required fields are marked accordingly.

  Marketing goal: Increase first-time purchases
  Brand description: GlowDrop
  Target audience: Women 22 to 38, skincare and wellness, UK and US
  Total budget: 5000
  Product / service price: 45
  Currency (e.g. GBP, EUR, USD) (optional, press Enter to skip): GBP
  Target market language (e.g. English, French): English
  Existing channels (optional, comma-separated, press Enter to skip): Organic TikTok only, 42k fol
lowers, reach declining
  Product type (B2B / B2C, press Enter to skip): B2C

Running pipeline…
```

Figure 2: Screenshot of Inputs

5.2 Agent's Processing Steps

5.2.1 Step 1 – Funnel Stage

The first step determines the overall strategic direction of the campaign. The agent first reads the marketing goal and the brand description to determine how to classify the campaign into one of the three funnel stages: Awareness, Consideration or Conversion. This classification is critical because the subsequent steps are determined by this output.

As shown in Figure 3 below, the agent identifies "increase first-time purchases" as a transactional goal, signalling that the purpose of this campaign is to drive purchases rather than build awareness and is therefore defined as a Conversion stage.

```
  Step 1 — Funnel Stage

  Stage : Conversion
  Why   : The goal "increase first-time purchases" targets a direct, measurable transactional action (a purchase), which defines the Conversion stage.
```

Figure 3: Screenshot of Step 1 – Funnel Stage

5.2.2 Step 2 – Selected Channels

Using the Conversion funnel stage from Step 1, together with the audience profile, budget, existing channels and product type, the agent then selects up to three advertising channels and assigns a pricing model that best suits the campaign objective.

For GlowDrop, the agent selects three channels – Google Search (CPC), Retargeting TikTok (CPA) and Meta Ads (CPC) – each chosen to maximise conversion for GlowDrop's specific audience and context, as shown in Figure 4.

```
Step 2 — Selected Channels

• Google Search [CPC]
  Captures high-intent searches from 22–38 women in the UK and US actively looking to buy skincare and wellness products, driving first-time purchases.
• Retargeting [CPA]
  Re-engages the existing 42k TikTok following and warm site visitors among these skincare-focused women to convert them into first-time buyers on a pay-per-acquisition basis.
• Meta Ads [CPC]
  Instagram's strong visual skincare and wellness communities let us target UK/US women 22–38 with shoppable, click-optimized ads that drive direct purchases.
```

Figure 4: Screenshot of Step 2 – Selected Channels

5.2.3 Step 3 – Budget Allocations

With the three channels selected, the agent then distributes the budget across based on each channel's conversion potential and role in the funnel. In our illustration, since GlowDrop's campaign example is conversion, Google Search receives the largest portion.

```
Step 3 — Budget Allocation

• Google Search: GBP 2,500.00 (50.0%)
  As the highest-intent channel capturing active purchase queries, Search leads the conversion stage ahead of Retargeting and Meta because it intercepts users already in buying mode.
• Retargeting: GBP 1,750.00 (35.0%)
  Retargeting is the strong secondary channel, re-engaging warm prospects who have already shown interest, complementing Search's new-intent capture with efficient CPA-priced reconversion.
• Meta Ads: GBP 750.00 (15.0%)
  Meta plays a supporting role here, sustaining lower-intent demand and feeding the audience pool that Search and Retargeting ultimately convert, so it receives the smallest share.
```

Figure 5: Screenshot of Step 3 – Budget Allocations

5.2.4 Step 4 – KPI Targets

Next, the agent calculates realistic performance targets for each channel based on the actual product price and the budget allocated per channel. In particular, CTR and CPA are derived from the funnel stage and channel characteristics while ROAS is calculated in Python as product price divided by CPA target, to eliminate risk of model calculation errors. Figure 6 shows the output for the mentioned GlowDrop example.

```
Step 4 — KPI Targets

• Google Search
  CTR 6.0%  |  CPA GBP 8.50  |  ROAS 5.29x
• Retargeting
  CTR 3.2%  |  CPA GBP 7.50  |  ROAS 6.00x
• Meta Ads
  CTR 1.5%  |  CPA GBP 12.50  |  ROAS 3.60x
```

Figure 6: Screenshot of Step 4 – KPI Targets

5.2.5 Step 5 – Keyword Generation

The agent generates five keywords in the target market language and matches it to the chosen funnel stage in Step 2. To ensure reliability, the agent validates that exactly five keywords are returned – first as an instruction to the AI model, followed by a Python check after the response is received. Figure 7 shows the 5 recommended keywords for the GlowDrop campaign.

```
Step 5 — Keywords

1. buy GlowDrop serum online
2. GlowDrop skincare first order discount
3. GlowDrop serum price UK
4. GlowDrop free trial sign up
5. order GlowDrop glow serum today
```

Figure 7: Screenshot of Step 5 – Keyword Generation

5.2.6 Step 6 – Campaign Brief

In the final step of the pipeline, the agent takes into consideration all outputs from Steps 1 to 5 and translates to a ready-to-use campaign brief in 3-5 sentences. Figure 8 shows the recommended campaign brief for the GlowDrop test example.

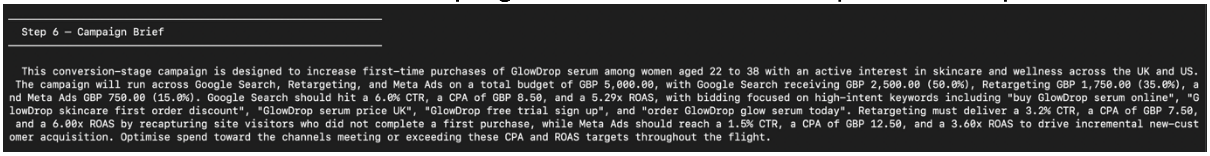

Figure 8: Screenshot of Step 6 – Campaign Brief

6. Business Value and Marketing Application

The Campaign Planning AI Agent is designed to initialize the entire marketing campaign planning workflow. As demonstrated in Section 5, the agent takes a set of structured inputs and produces a complete campaign package in a few minutes, therefore providing marketing teams with a ready-to-use campaign brief. Rather than replacing the creative and execution aspects of marketing, the agent acts as an accelerator that bridges ideation and implementation, enabling marketing teams to work more effectively. In particular, the integration of the agent into the marketing campaign planning process delivers three key business benefits: speed, consistency, and scalability.

6.1 Business Value 1: Speed

One of the most significant benefits of the AI agent is the time savings. As outlined in Section 2, marketers spend around four hours of manual work. The agent can help alleviate such manual work by eliminating the most time-intensive preparatory stages such as conducting initial brainstorming sessions, researching the channels, allocating budgets etc., compressing into a process that requires two minutes only.

As shown in the GlowDrop demonstration in Section 5, the agent produces the key campaign parameters – channel select, budget allocation, KPI targets, target audiences and keywords in a ready-to-use campaign brief that is tailored to the specific channel selected. This allows teams to redirect their time towards other high-value activities such as refining campaign strategy, developing creative and managing stakeholder engagement, leading to improved productivity and a significantly shorter campaign planning cycle.

6.2 Business Value 2: Consistency

The agent also helps ensure consistency by applying a standardised framework across all campaigns. Rather than relying solely on subjective judgement or individual experience (as mentioned in Section 2), the agent evaluates every set of business inputs through the same structured six-step pipeline, therefore reducing variation in planning quality and design across campaigns and team members.

It is also important to note that consistency does not come at the cost of tailored outputs. As illustrated in Section 3, the agent adapts its outputs to the specific inputs and business context provided – a B2C skincare brand focused on conversion will thus receive a fundamentally different brief from a B2B software company targeting awareness. The planning methodology remains consistent while the recommendations remain specific to each campaign needs.

6.3 Business Value 3: Scalability

Incorporating the agent increases the volume of campaigns a marketing team can handle as well, without a proportional increase in resources needed. This is particularly valuable during peak promotional periods such as seasonal campaigns or product launches.

By automating the generation of campaign recommendations and planning outputs, the agent provides a scalable planning capability that is applicable across multiple brands, different products or market segments. This enables organizations to maintain planning quality as business volume grows, ensuring that the marketing function supports rather than becoming a bottleneck to commercial activities.

6.4 Applying to the Business Environment

This agent is particularly relevant for organizations that require efficient and repeatable campaign planning processes but have limited resources available for marketing activities.

Smaller marketing agencies can benefit from the agent by reducing the effort required to develop campaign efforts for multiple clients. This enables teams to manage a larger portfolio of campaigns while ensuring a structured and reliable planning approach. Likewise, freelance marketers also stand to benefit greatly with increased capacity and improved consistency across their work.

In larger organizations, the AI Agent is also able to generate substantial value by supporting internal marketing in providing a standardized planning framework across multiple brands, products or business units, ensuring a similar tone in campaign briefs produced across. This helps to ensure that the campaign briefs produced are largely aligned and scalable as marketing operations grow.

6.5 Quantifying the Business Impact

Assume a marketing team produces 8 campaign briefs each month and stands to save 2 hours with the use of the Campaign Planning AI Agent. There is a total of 3 marketing personnels in the team, and each earns an average of €50 per hour.

<u>Per Marketing Personnel</u>
Total hours saved across brief/ month = 2 hours x 8 briefs = 16 hours
Translating hours saved to monetary value/ month = 16 hours x €50/ hour = €800

<u>3 Marketing Personnel</u>
Cost savings for 3 marketing personnels = €800 x 3 = €2,400
Annual cost savings = €2,400 x 12 = €28,800

As shown in the calculations above, a marketing team can expect to realise €28,800 from their daily operations and reallocate this quantified timing to other important tasks, therefore improving the overall productivity of the team.

6.6 Summary of Business Impact

Incorporating AI Agents in the marketing campaign process has the potential to fundamentally transform how marketing decisions are made and how campaign planning activities are performed. These AI Agents act as collaborative tools that accelerate planning tasks and enhance the bridge between ideation and implementation.

7. Reflection

7.1 Strengths

Beyond what a traditional AI tool might provide, the Campaign Planning AI Agent exhibits a number of significant advantages. Instead of just producing text, the agent takes deliberate decisions at every stage, from budget allocation to channel selection, and offers a rationale for each action. Its transparency in reasoning sets it apart from a prompt-and-response system and makes it a true agent.

Critical requirements like budget summing to exactly 100% and generating precisely five keywords are assured regardless of model variability because constraints are simultaneously enforced at two levels: within the prompt instructions and within the Python code itself.

Risks of arithmetic errors in the output are also almost eliminated as financial measures like ROAS and budget normalization are calculated deterministically in code rather than being left to the language model. The spec-driven development process produced clean, maintainable, and well-structured code, resulting in a brief that a media buyer can use right away without additional tweaking.

7.2 Limitations

However, the agent has limitations that are worth acknowledging honestly. Firstly, an Anthropic API key is needed to run the entire pipeline, which adds a dependency and expense for teams wishing to use it at scale.

Secondly, since the agent's KPI targets are based on broad benchmark ranges rather than real-time platform data, they represent plausible industry estimations rather than the state of the market.

Third, the quality of the output is directly tied to the quality of the input, hence any incomplete or ambiguous brand descriptions will result in suggestions that are less accurate and useful.

Fourth, in some runs the agent recommends channel strategies such as Retargeting rather than specific advertising platforms. In the GlowDrop example, this reflects a deliberate use of the brand's existing 42,000 TikTok followers as a warm audience to retarget – leveraging established brand equity instead of building reach from scratch. In practice, a media buyer would configure this retargeting campaign on the most appropriate platform, reflecting an area where human review remains essential.

Lastly, because the agent lacks memory in between sessions – every run begins completely from scratch and the agent does not learn from or build upon prior campaigns.

7.3 Risks

The agent might offer suboptimal channel recommendations for extremely specialized or niche markets where the general logic of media planning may not be the most suitable. Budget splits are based on general strategic reasoning that may need to be modified in light of platform-specific details that the agent is currently unable to take into consideration. Most significantly, there is a genuine risk of launching campaigns based on assumptions that do not accurately reflect the business context if the agent's output is relied upon excessively without human scrutiny.

7.4 Improvements

Connecting the agent to real data from the Google Ads and Meta APIs would be the most significant upgrade going forward, enabling it to use actual CPC and CPM numbers instead of benchmarks. The agent would be able to improve its suggestions over time if it included a feedback loop that takes into account the outcomes of previous campaigns. Multi-language support would increase the tool's utility for teams working in non-English markets, and a simple web interface would make the tool accessible to marketers who are not comfortable working in a terminal environment.

7.5 Real-World Marketing Application

When considering how the agent would function in a real marketing environment, its value lies in accelerating the most time-intensive stages of campaign planning – the research, analysis and structuring. Rather than replacing existing workflows, the agent is best positioned as the first step in a campaign planning cycle, producing a structured brief that a marketing manager can hand directly to a media buyer, with creative direction layered in separately. As demonstrated in Section 5, this process can be completed within minutes instead of hours.

The agent therefore works best as a collaborative tool rather than a standalone solution. Critical aspects of campaign planning that require human judgement such as stakeholder alignment remain the responsibility of the marketer. The agent's primary value lies in establishing a robust strategic foundation before these discussions take place, enabling teams to focus on higher-level decision-making rather than routine planning tasks. As discussed in Section 6.4, this capability is particularly beneficial for smaller marketing teams, freelancers and large organizations managing multiple brands or campaigns simultaneously.

8. Conclusion

The Campaign Planning Agent shows that when AI is created with structure, constraints, and a clear business goal in mind, it can go well beyond text production. Without compromising the logic or context-sensitivity necessary for effective campaign planning, the agent reduces what typically takes a marketing team two to three hours into less than two minutes by automating the six-step process of funnel classification, channel selection, budget allocation, KPI calculation, keyword generation, and brief writing.

The agent's combination of deterministic computing and AI judgment, in addition to its speed, is what makes it so intriguing. While Python code enforces the constraints that cannot be left to chance, like making sure the budget always adds up to exactly 100% and that KPI targets are derived from actual product prices rather than generic industry averages, the language model handles the decisions that require interpretation, like determining the appropriate funnel stage or explaining why a specific channel suits a given audience. The result is dependable enough to be given straight to a media buyer because of this separation of responsibilities between the model and the code.

The GlowDrop demonstration in section 5 exemplifies this effectively. The agent created a robust conversion-stage strategy that prioritized Google Search for high-intent purchase queries, used Retargeting to re-engage skincare interested shoppers, and assigned Meta Ads a supporting role given a brand with an existing 42k-follower organic presence, a defined monthly budget, and a clearly defined audience of women aged 22 to 38 interested in skincare and wellness across the UK and US.

The business case is just as obvious. As mentioned in section 6, a three-person marketing team that produces eight briefs a month might save over €28,800 annually in planning time alone. In addition to cost reductions, the agent offers scalability that doesn't necessitate proportionate personnel expansion and consistency across campaigns and team members. For smaller firms, independent contractors, and larger companies handling several brands at once, these are significant benefits.

This study has acknowledged the agent's limits, which are true. Its output is only as powerful as the inputs it takes; it lacks memory between sessions, and it does not connect to real-time platform data. The method still requires human evaluation, especially for campaigns with unique budget structures or specialist audiences. It is advisable to think of the agent as a place to start that speeds up and standardizes planning rather than as a substitute for marketing knowledge.

There is a clear way ahead. Live API connections to Google Ads and Meta would replace benchmark estimates with real market data. A feedback loop incorporating past campaign results would allow the agent to improve its recommendations over time. A simple web interface would make it accessible to non-technical users.
Taken together, the Campaign Planning Agent is a practical, well-reasoned tool that addresses a genuine and recurring pain point in marketing operations. It does not attempt to replace the marketer. It gives them a better, faster, and more consistent place to start.

References

HubSpot. (2024). *State of marketing report 2024*. https://www.hubspot.com/state-of-marketing